

Monotone submodular maximization

Cardinality constraints

$f: 2^n \rightarrow \mathbb{R}_{\geq 0}$ monotone, submodular, normalized ($f(\emptyset) = 0$)

$\max \{f(S) : S \subseteq [n], |S| \leq k\}$ eg. max k-cover

Thm: $(1 - \frac{1}{e})$ -approx

Greedy: $A \leftarrow \emptyset$

pick $a \in \arg\max f(a|A)$

$A \leftarrow A \cup \{a\}$

Pf: Assume $Alg = \{a_1, \dots, a_k\}$

$A_i = \{a_1, \dots, a_i\}$

$$f(a_i | A_{i-1}) \geq \frac{1}{k} \sum_{o \in OPT} f(o | A_{i-1}) \stackrel{\text{(submod)}}{\geq} \frac{1}{k} f(o_{PT} | A_{i-1}) \stackrel{\text{(monotone)}}{\geq} \frac{1}{k} (f(o_{PT}) - f(A_{i-1}))$$

$$\Rightarrow f(A_i) - f(A_{i-1}) \geq \frac{1}{k} (f(o_{PT}) - f(A_{i-1}))$$

(continuous) $y_i = f(A_i)$, $T = [0, 1]$, $dt = \frac{1}{k}$

$$\frac{dy}{dt} \geq (f(o_{PT}) - y)$$

$$\Rightarrow \int_0^{f(Alg)} \frac{1}{f(o_{PT}) - y} dy \geq \int_0^1 dt$$

$$\Rightarrow -\ln(f(o_{PT}) - f(Alg)) + \ln(f(o_{PT})) \geq 1$$

$$\Rightarrow \frac{f(o_{PT})}{f(o_{PT}) - f(Alg)} \geq e$$

$$f(Alg) \geq (1 - \frac{1}{e}) \cdot f(o_{PT})$$

(discrete) $f(\text{OPT}) - f(A_i) \leq \alpha (f(\text{OPT}) - f(A_{i-1}))$

$$(1-\alpha) \cdot f(\text{OPT}) \leq f(A_i) - \alpha \cdot f(A_{i-1})$$

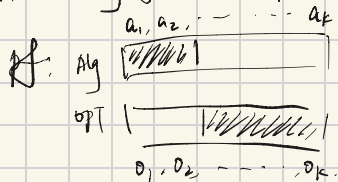
$$\alpha \triangleq 1 - \frac{1}{e}$$

$$\Rightarrow f(\text{OPT}) - f(\text{Alg}) \leq (1 - \frac{1}{e})^k \cdot f(\text{OPT}) \leq \frac{1}{e} \cdot f(\text{OPT})$$

$$\Rightarrow f(\text{Alg}) \geq (1 - \frac{1}{e}) \cdot f(\text{OPT})$$

Matroid Constraints

Thm: Greedy gets $\frac{1}{e}$ -approx.



OK: Alg picks a basis (monotonicity is important)

$$A_k = \{a_1, \dots, a_k\}$$

$$O_k = \{o_1, \dots, o_k\}$$

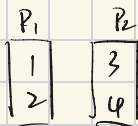
$$A_k + (\text{OPT} \setminus O_k) \in \mathcal{B}(M)$$

$$f(\text{Alg}) = \sum_{i=1}^k f(a_i | A_{i-1}) \geq \sum_{i=1}^k f(o_i | A_{i-1}) \geq \sum_{i=1}^k f(o_i | \text{Alg}) \geq f(\text{OPT} | \text{Alg})$$

(submod) (submod)

$$\Rightarrow f(\text{Alg}) \geq \frac{1}{e} \cdot f(\text{OPT})$$

tight example: partition matroid constraints



universe $U = \{x, y, z\}$

Coverage function: $S_1 = \{x\}$ $S_3 = S_4 = \{x, z\}$

$S_2 = \{y\}$

(weighted coverage): $w: U \rightarrow \mathbb{R}_{\geq 0}$

$$f(F) = w\left(\bigcup_{e \in F} e\right) \quad \forall F \subseteq E$$

Nonmonotone submodular maximization, (nonnegative, not normalized)

1. unconstrained

eg: cut function: $f(U) = |S(U)| \quad \forall U \subseteq V$ 
 directed cut function: $f(U) = |S^+(U)| \quad \forall U \subseteq V$

cut: $\frac{1}{2}$ -approx by random sample. directed cut: $\frac{1}{4}$ -approx

Thm: Random sample gets $\frac{1}{4}$ -approx for nonmonotone submod
 $\frac{1}{2}$ -approx for symmetric nonmonotone submod

Alg: Independently sample $y_i = \begin{cases} 1 & \text{w.p. } \frac{1}{2} \\ 0 & \text{w.p. } \frac{1}{2} \end{cases}$, $S = \{i: x_i = 1\}$.

"Multilinear extension" $x \in [0, 1]^n$, let $R(x) \subseteq [n]$ be a random set where every i is sampled independently with prob. x_i $f(x) = \mathbb{E}(f(R(x)))$

$$f(x) = \sum_{S \subseteq [n]} \prod_{i \in S} x_i \prod_{i \notin S} (1 - x_i) \cdot f(S)$$

Lemma: f is concave in direction $d \geq 0$



$$\frac{\partial f}{\partial x_i} = f(x \vee e_i) - f(x \wedge (1 - e_i)) = \mathbb{E}(f(R(x) \vee i)) - \mathbb{E}(f(R(x) \setminus i))$$

$\rightarrow$ multilinear

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial f(x \vee e_i)}{\partial x_j} - \frac{\partial f(x \wedge (1 - e_i))}{\partial x_j}$$

$$= f(x \vee e_i \vee e_j) - f(x \vee e_i \wedge \bar{e}_j) - (f(x \wedge \bar{e}_i \vee e_j) - f(x \wedge \bar{e}_i \wedge \bar{e}_j))$$

$$= \mathbb{E}(f(R \vee i \vee j) - f(R \vee i \setminus j) - f(R \setminus i \vee j) + f(R \setminus i \setminus j))$$

$$(\text{if } j \neq i) \leq 0$$

$$\nabla^2 f = \begin{pmatrix} 0 & & \\ & 0 & \\ - & & \ddots \\ & & & 0 \end{pmatrix}$$

$$d \in \mathbb{R}_+^n, t \in \mathbb{R}$$

$$g(t) = f(x + td)$$

$$g'(t) = d^T \nabla^2 f d \leq 0$$



pf of Thm:

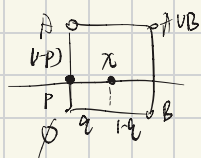
claim: For $A \subseteq [n]$, $0 \leq p \leq 1$, denote by $A(p)$ the random set where every $i \in A$ is sampled w.p. p independently. then

$$\mathbb{E}[f(A(p))] = \mathbb{E}[f(\mathbb{1}_A \cdot p)] = p \cdot f(A) + (1-p) \cdot f(\emptyset)$$

claim: For $p, q \in [0, 1]$, $\mathbb{E}[f(A(p) \cup B(q))] \geq (1-p)(1-q) \cdot f(\emptyset) + p(1-q) \cdot f(A) + (1-p)q \cdot f(B) + pq \cdot f(A \cup B)$

(let $g(B) := g(B \cup A(p))$)

$$\begin{aligned} \mathbb{E}[f(A(p) \cup B(q))] &\geq \mathbb{E}[q \cdot f(B \cup A(p)) + (1-q) \cdot f(A(p))] \\ &\geq q(p \cdot f(A \cup B) + (1-p) \cdot f(A)) + (1-q)(p \cdot f(A) + (1-p) \cdot f(\emptyset)) \\ &= \text{above} \end{aligned}$$



let $A = [n/2]$, $B = [n] \setminus [n/2] \Rightarrow A \upharpoonright q = A(\frac{1}{2}) \cup B(\frac{1}{2})$

$$\begin{aligned} \Rightarrow \mathbb{E}[f(A \upharpoonright q)] &\geq \frac{1}{4} f(\emptyset) + \frac{1}{4} f([n/2]) + \frac{1}{4} f([n/2]) + \frac{1}{4} f([n]) \\ &\geq \frac{1}{4} f([n/2]) + \frac{1}{4} f([n/2]) \left\{ \begin{array}{l} \geq \frac{1}{4} f([n/2]) \\ = \frac{1}{2} f([n/2]) \text{ symmetric } f \end{array} \right. \end{aligned}$$

Double greedy:

deterministic alg:

$$A \leftarrow \emptyset, \bar{B} \leftarrow \emptyset$$

For $i = 1 \rightarrow n$

$$\alpha \leftarrow f(e_i | A)$$

$$\beta \leftarrow f(e_i | \bar{B})$$

$$\text{if } \alpha \geq \beta : A \leftarrow A + e_i$$

$$\text{else } \bar{B} \leftarrow \bar{B} + e_i$$

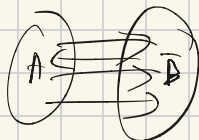
Output A (or equivalently $\bar{B}$)

$$E = \{e_1, \dots, e_n\}$$

$$\text{let } \bar{f}(S) := f(E \setminus S)$$

Goal: A maximizes f

$\bar{B}$ maximizes $\bar{f}$ ($\Leftrightarrow B$ maximizes f)



$$f(B + e) - f(B) = \bar{f}(\bar{B} \cup e) - \bar{f}(\bar{B}) = \bar{f}(e | \bar{B})$$

Thm: $\frac{1}{2}$ -approx.

randomized alg.

$$A \in \mathcal{P}, \bar{B} \in \mathcal{P}$$

For $i=1, \dots, n$

$$\alpha_i \leftarrow \max\{f(e_i|A), 0\}$$

$$\beta_i \leftarrow \max\{\bar{f}(e_i|\bar{B}), 0\}$$

If $\alpha + \beta \neq 0$

$$\begin{cases} A \leftarrow A + e_i \text{ w.p. } \frac{\alpha_i}{\alpha_i + \beta_i} \\ \bar{B} \leftarrow \bar{B} + e_i \text{ w.p. } \frac{\beta_i}{\alpha_i + \beta_i} \end{cases}$$

else: $A \leftarrow A + e_i$

Output A (or equivalently B)

Analysis:

Lemma: Define $D_i := \text{OPT} \cup A_i \setminus \bar{B}_i$, then,

$$\mathbb{E}(f(D_{i+1}) - f(D_i)) \leq \frac{1}{2} \mathbb{E}(f(A_i) - f(A_{i-1}) + \bar{f}(\bar{B}_i) - \bar{f}(\bar{B}_{i-1})) \quad (*)$$

Then: Randomized double greedy gets $\frac{1}{2}$ -approx

pf: Since $D_0 = \text{OPT}$, $D_n = A$

$$\stackrel{\text{lemma}}{\Rightarrow} \mathbb{E}(f(\text{OPT}) - f(A)) \leq \frac{1}{2} \mathbb{E}(f(A) + \bar{f}(\bar{B})) = \mathbb{E}(f(A))$$

$$\Rightarrow \mathbb{E}(f(A)) \geq f(\text{OPT})$$

pf of lemma:

• $\alpha_i + \beta_i \neq 0$

$$\text{claim: } \mathbb{E}[f(A_i) - f(A_{i-1}) + \bar{f}(\bar{B}_i) - \bar{f}(\bar{B}_{i-1})] = \frac{\alpha_i^2 + \beta_i^2}{\alpha_i + \beta_i}$$

Condition on $A_{i-1}, \bar{B}_{i-1}$

pf: $\alpha_i, \beta_i \geq 0 \quad \checkmark$

Suppose $\alpha_i = 0, \beta_i > 0 \quad \checkmark \quad \square$

Case 1: $e_i \notin \text{OPT}$

$$\mathbb{E}(f(O_{i-1}) - f(O_i)) = \mathbb{P}(e_i \in A_i) \cdot (-f(e_i | O_{i-1}))$$

$$\begin{aligned} A_i & \text{ (diagram: a rectangle with a shaded region labeled } A_i \text{ and } \overline{B}_i \text{)} \\ \overline{B}_i & \text{ (diagram: a rectangle with a shaded region labeled } \overline{B}_i \text{)} \end{aligned} \leq \frac{\alpha_i}{\alpha_i + \beta_i} \cdot (-f(e_i | B_{i-1} - e_i)) \quad (B_i = \overline{B}_i)$$

$$O_i = \text{opt} \cup A_i \setminus \overline{B}_i \leq \frac{\alpha_i \beta_i}{\alpha_i + \beta_i}$$

Case 2: $e_i \in \text{opt}$

$$\begin{aligned} \mathbb{E}(f(O_{i-1}) - f(O_i)) &= \mathbb{P}(e_i \in \overline{B}_i) \cdot f(e_i | O_{i-1} - e_i) \\ &\leq \frac{\beta_i}{\alpha_i + \beta_i} f(e_i | A_{i-1}) \\ &\leq \frac{\alpha_i \beta_i}{\alpha_i + \beta_i} \end{aligned}$$

$$\alpha_i^2 + \beta_i^2 \geq 2\alpha_i \beta_i \Rightarrow \text{lemma}$$

• $\alpha_i + \beta_i = 0$

Since $A_{i-1} \cap \overline{B}_i = \emptyset$, $A_{i-1} \subseteq B_{i-1} - e_i$

$$\Rightarrow f(e_i | A_{i-1}) \geq f(e_i | B_{i-1} - e_i) \Rightarrow f(e_i | A_{i-1}) + \bar{f}(\bar{e}_i | \overline{B}_i) \geq 0$$

$$\alpha_i + \beta_i = 0 \Rightarrow f(e_i | A_{i-1}) = \bar{f}(\bar{e}_i | \overline{B}_i) = 0 \Rightarrow \text{RHS of } (*) = 0$$

$$\text{LHS} = \begin{cases} -f(e_i | O_{i-1}) \leq \beta_i = 0 & \text{if } e_i \notin \text{opt} \\ 0 & \text{if } e_i \in \text{opt} \end{cases}$$



Continuous greedy:

Recall multilinear extension. $\bar{F}(x) = \mathbb{E}(F(R(x)))$

where $R(x)$ is a random set where $i \in [n]$ is sampled w.p. x_i

① $\bar{F}$ is monotone if f is monotone

② $\bar{F}$ is concave in direction $d \in \mathbb{R}_{\geq 0}^n$

③ $\bar{F}$ is convex in direction $e_i - e_j$

Goal: $\max_{x \in P} f(x)$, where $P \subseteq [0,1]^n$ is some polytope

Alg: Fix some step size $S \ll 1$.

Initialize $x = 0$, $t = 0$

While $t < 1$.

$$v^* \leftarrow \operatorname{argmax}_{v \in P} \langle v, \nabla f(x) \rangle$$

$$x \leftarrow x + S \cdot v^*, \quad t \leftarrow t + S$$

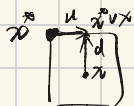
Return x

Remark: v^* is an extreme point of P .

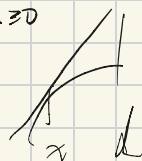
Final solution $x = \sum_i S_i v_i^*$ is a convex combination of extreme points of $P \Rightarrow x \in P$.

Thm: Continuous greedy gets $1 - \frac{1}{e}$

Analysis: Let $x^* = 1_{\text{opt}}$, $d := (x^* \vee x) - x = (x^* - x) \vee 0 \Rightarrow d \in \mathbb{R}^n_{\geq 0}$



$$x^* \geq d$$



$$\begin{aligned} \frac{d f(x(t))}{dt} &= \langle \nabla f(x), \frac{dx}{dt} \rangle = \langle \nabla f(x), v^* \rangle \stackrel{\text{(monotonicity)}}{\geq} \langle \nabla f(x), x^* \rangle \geq \langle \nabla f(x), d \rangle \\ &\stackrel{\text{(concavity)}}{\geq} f(x + d) - f(x) \stackrel{\text{(monotonicity)}}{\geq} f(x^*) - f(x) = f_{\text{opt}} - f(x(t)) \end{aligned}$$

$$\begin{aligned} \int_0^t \frac{d(e^{t'} f(x(t')))}{dt'} dt' &\geq \int_0^t e^{t'} f_{\text{opt}} dt' \Rightarrow e^t f(x(t)) \geq (e^t - 1) \cdot f_{\text{opt}} \\ &\Rightarrow f(x(t)) \geq (1 - e^{-t}) \cdot f_{\text{opt}} \end{aligned}$$

Nonmonotone submodular maximization (constrained)

$\max_{x \in P} f(x)$, where P is some downward-closed polytope.

Continuous greedy gets $(1 - \ln 2) \approx 0.307$.

Measured continuous greedy

Alg: Fix some step size $S < 1$.

Initialize $x = 0, t = 0$

While $t < 1$:

$$v^* \leftarrow \operatorname{argmax}_x \langle v, \nabla f(x) \rangle : v \in P, v \leq 1 - x$$

$$x \leftarrow x + S \cdot v^*, t \leftarrow t + S$$

Return x

Thm: Measured greedy gets $(1 - \frac{1}{e})$ for monotone, $\frac{1}{e^{1.368}}$ for nonmonotone

pf: monotone: $d = (x^* - x) \vee 0 \leq 1 - x \quad d \leq x^* \Rightarrow d \in P$

$$\Rightarrow \langle \nabla f(x), v^* \rangle \geq \langle \nabla f(x), d \rangle$$

nonmonotone:

lemma: if $x_i \in P \leq 1 \quad \forall i \in [n]$, then $f(x \vee x^*) \geq (1-p) f(x^*)$

pf: let $u := (x \vee x^*) - x^* = (x - x^*) \vee 0 \in \mathbb{R}_{\geq 0}^d$

$$g(\lambda) := x^* + \lambda \cdot u$$

$$g(\frac{1}{p}) = x^* + \frac{1}{p} \cdot ((x - x^*) \vee 0) = \left(\frac{1}{p} x + (1 - \frac{1}{p}) x^* \right) \vee x^* \leq \left(\frac{1}{p} x \right) \vee x^* \leq 1$$

$$f(x \vee x^*) = g(1) = g((1-p) \cdot 0 + p \cdot \frac{1}{p}) \geq \underline{(1-p) \cdot g(0)} + p \underline{g(\frac{1}{p})} \geq \underline{(1-p) \cdot f(x^*)}$$

lemma: $x_i(t) \leq 1 - e^{-t}$, $\forall i \in [n]$.

pf: $x'_i(t) = v_i \leq 1 - x_i(t) \Rightarrow x_i(t) \leq 1 - e^{-t}$

pf of Thm (nonmonotone): $\frac{d f(x(t))}{dt} \geq f(x \vee x^*) - f(x) \geq e^{-t} \cdot f(x^*) - f(x(t))$

$$\Rightarrow \int_0^t \frac{d(e^{-t} \cdot f(x(t)))}{dt} dt \geq \int_0^t f(x^*) dt \Rightarrow e^{-t} f(x(t)) \geq f(x^*)$$

i.e. $f(x(t)) \geq t \cdot e^{-t} \cdot f(\operatorname{opt})$

eg, $P = \{x : x_1 + \dots + x_n \leq k\}$

P matroid polytope.